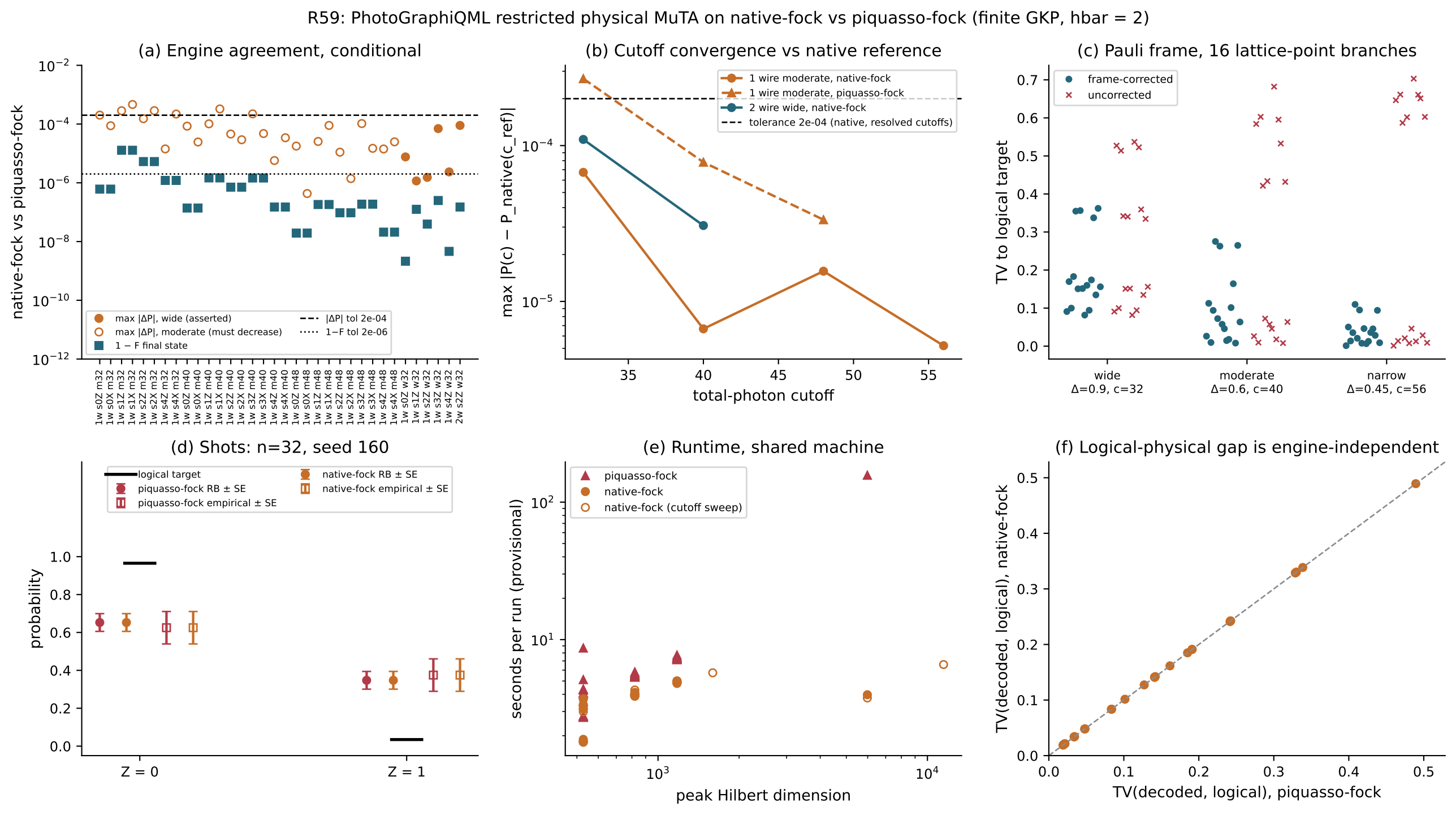
R59: PhotoGraphiQML restricted physical MuTA on native-fock vs piquasso-fock (finite GKP, hbar = 2)
(a) Engine agreement, conditional
native-fock vs piquasso-fock
max |ΔP|, wide (asserted)
max |ΔP|, moderate (must decrease)
1 − F final state
|ΔP| tol 2e-04
1−F tol 2e-06
(b) Cutoff convergence vs native reference
max |P(c) − P_native(c_ref)|
total-photon cutoff
1 wire moderate, native-fock
1 wire moderate, piquasso-fock
2 wire wide, native-fock
tolerance 2e-04 (native, resolved cutoffs)
(c) Pauli frame, 16 lattice-point branches
TV to logical target
frame-corrected
uncorrected
wide
Δ=0.9, c=32
moderate
Δ=0.6, c=40
narrow
Δ=0.45, c=56
(d) Shots: n=32, seed 160
probability
logical target
piquasso-fock RB ± SE
piquasso-fock empirical ± SE
native-fock RB ± SE
native-fock empirical ± SE
Z = 0
Z = 1
(e) Runtime, shared machine
seconds per run (provisional)
peak Hilbert dimension
piquasso-fock
native-fock
native-fock (cutoff sweep)
(f) Logical-physical gap is engine-independent
TV(decoded, logical), native-fock
TV(decoded, logical), piquasso-fock